

No.	Attribute description	Abbrev.
x_1	Magnitude in the range 2-6 (in ML)	Magnitude
x_2	Depth in the range 0-200 (in km)	Depth
x_3	Latitude in the range 43.7 to 48.2 (in Degrees)	Latitude
x_4	Longitude in the range 20.7 to 29.7 (in Degrees)	Longitude
y	The strength of the earthquake	Earthquake level

Table 1: Description of the attributes of the Earthquakes in Romania dataset used in this exam. The dataset contains all the earthquakes that originated in Romania since late 2012 until today. It includes the magnitude as measured by Richter local magnitude (ML), the depth of the earthquake, the latitude and longitude. The dataset used here consists of $N = 3714$ observations and the attribute y is discrete such that $y = 1$ (corresponds to Low strength), and $y = 2$ (corresponds to High strength).

Question 1. The main dataset used in this exam is the Earthquakes in Romania dataset¹ described in Table 1. We consider the four attributes of the Earthquakes in Romania dataset in Table 1. A scatter plot matrix for these attributes is given in Figure 1.

Which one of the following matrices represents the empirical correlation matrix for these attributes?

A. $\hat{C} \approx \begin{bmatrix} 1.0 & 0.48 & 0.98 & 0.21 \\ 0.48 & 1.0 & 0.02 & 0.46 \\ 0.98 & 0.02 & 1.0 & 0.07 \\ 0.21 & 0.46 & 0.07 & 1.0 \end{bmatrix}$

B. $\hat{C} \approx \begin{bmatrix} 1.0 & -0.48 & 0.02 & 0.21 \\ -0.48 & 1.0 & 0.02 & 0.46 \\ 0.02 & 0.02 & 1.0 & 0.07 \\ 0.21 & 0.46 & 0.07 & 1.0 \end{bmatrix}$

C. $\hat{C} \approx \begin{bmatrix} 1.0 & 0.48 & 0.02 & 0.21 \\ 0.48 & 1.0 & 0.02 & 0.46 \\ 0.02 & 0.02 & 1.0 & 0.07 \\ 0.21 & 0.46 & 0.07 & 1.0 \end{bmatrix}$

D. $\hat{C} \approx \begin{bmatrix} 1.0 & 0.02 & 0.02 & 0.21 \\ 0.02 & 1.0 & 0.02 & 0.46 \\ 0.02 & 0.02 & 1.0 & 0.07 \\ 0.21 & 0.46 & 0.07 & 1.0 \end{bmatrix}$

E. Don't know.

Solution 1. The correct option is C. In the other options there are mistakes as high-correlation in the

¹Dataset obtained from <https://www.kaggle.com/datasets/stefancomanita/earthquakes-in-romania-from-2012-to-2024>

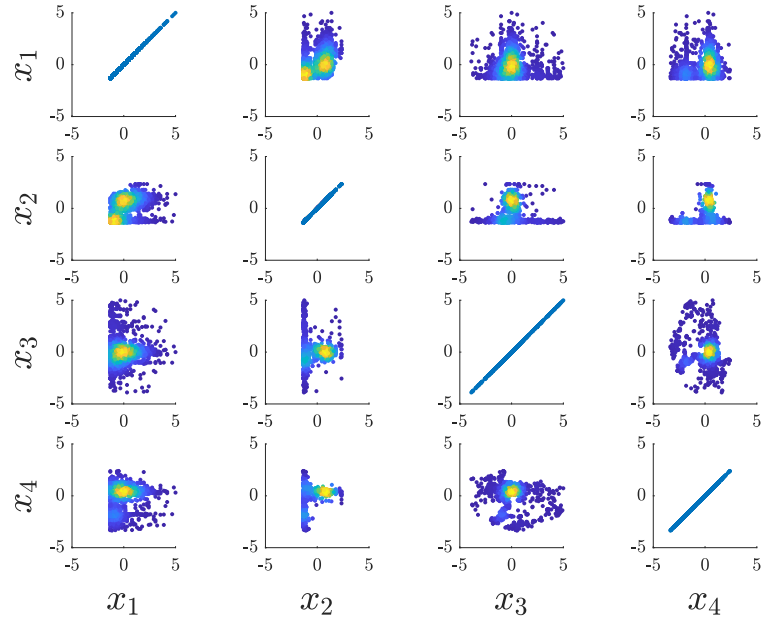


Figure 1: The scatter plot matrix for the attributes x_1 , x_2 , x_3 and x_4 of the Earthquakes in Romania dataset of Table 1. In the off-diagonal panels, the color represents the associated density.

matrix, but small-correlation in the scatter plot, difference between negative/positive correlation between the matrix and the scatter plot, etc.

Question 2. We consider the histograms of the four attributes in Figure 2. Match the histograms to the attributes by using the information regarding the distributions of the points from Figure 1.

Which one of the following statements is *correct*?

- A. Histogram 1: x_3 , Histogram 2: x_4
Histogram 3: x_2 , Histogram 4: x_1 .
- B. **Histogram 1: x_1 , Histogram 2: x_4**
Histogram 3: x_3 , Histogram 4: x_2 .
- C. Histogram 1: x_2 , Histogram 2: x_1
Histogram 3: x_3 , Histogram 4: x_4 .
- D. Histogram 1: x_3 , Histogram 2: x_2
Histogram 3: x_4 , Histogram 4: x_1 .
- E. Don't know.

Solution 2. We consider the density plots per column (or per row). If we exclude the distribution along the y -axis we can get the density per attribute along the x -axis. Using this knowledge, the option that matches the histograms to the density plots is B.

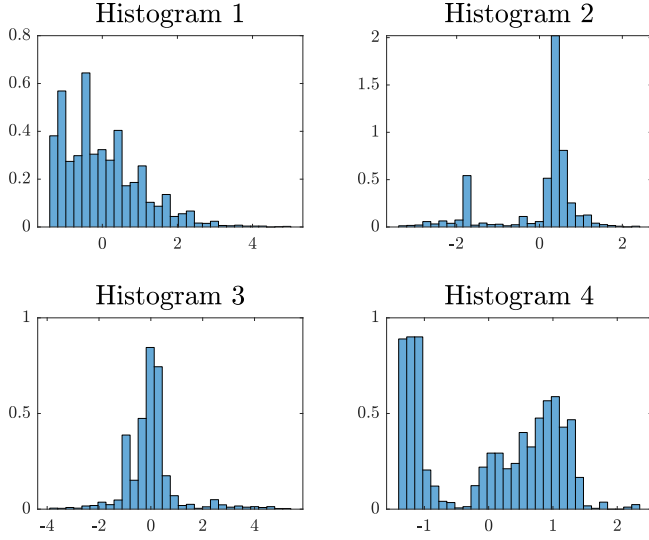


Figure 2: Histograms of the attributes in which their order has been permuted.

Question 3. Consider the PCA analysis for the Earthquakes in Romania dataset, in particular the SVD decomposition of the centered data matrix $\tilde{\mathbf{X}}$, and the corresponding matrix

$$\mathbf{V} = \begin{bmatrix} 0.54 & -0.13 & 0.68 & -0.48 \\ 0.65 & -0.08 & 0.0 & 0.76 \\ 0.07 & 0.98 & 0.16 & 0.04 \\ 0.53 & 0.09 & -0.72 & -0.44 \end{bmatrix}.$$

Which one of the following statements is *correct*?

- A. An observation with a high value of x_1 (**Magnitude**), a high value of x_2 (**Depth**), and a high value of x_4 (**Longitude**) will typically have a negative value of the projection onto principal component number 1.
- B. An observation with a low value of x_1 (Magnitude), and a high value of x_4 (Longitude) will typically have a negative value of the projection onto principal component number 3.**
- C. An observation with a low value of x_3 (**Latitude**) will typically have a positive value of the projection onto principal component number 2.
- D. An observation with a low value of x_1 (**Magnitude**), a high value of x_2 (**Depth**), and a low value of x_4 (**Longitude**) will typically have a negative value of the projection onto principal component number 4.
- E. Don't know.

Solution 3. The correct answer is B. Focusing on the correct answer, note the projection onto principal component $\mathbf{v}_3$ (i.e. column three of $\mathbf{V}$) is

$$\mathbf{b}_3 = \mathbf{x}^\top \mathbf{v}_3 = \begin{bmatrix} x_1 & x_2 & x_3 & x_4 \end{bmatrix} \begin{bmatrix} 0.68 \\ 0.0 \\ 0.16 \\ -0.72 \end{bmatrix}$$

(we use these attributes since these were selected for the PCA). It is now a simple matter of observing that for this number to be (relatively large) and negative, this occurs if x_1, x_4 has large magnitude and the sign convention given in option B.

Question 4. We consider the dataset in Table 2 with the binary attributes x_j for $j = 1, \dots, 4$. We construct a Bayes classifier relying on the attributes x_2 and x_4 .

Which one of the following probabilities is *correct*?

- A. $p(y = 2 \mid x_2 = 1, x_4 = 1) \approx 0.67$**
- B. $p(y = 2 \mid x_2 = 1, x_4 = 1) \approx 0.5$
- C. $p(y = 2 \mid x_2 = 1, x_4 = 1) \approx 0.39$
- D. $p(y = 2 \mid x_2 = 1, x_4 = 1) \approx 0.92$
- E. Don't know.

	x_1	x_2	x_3	x_4	y
$\mathbf{o}_1$	0	0	0	0	1
$\mathbf{o}_2$	0	0	1	1	1
$\mathbf{o}_3$	1	1	0	0	1
$\mathbf{o}_4$	0	1	1	1	1
$\mathbf{o}_5$	1	1	0	1	2
$\mathbf{o}_6$	0	0	0	0	2
$\mathbf{o}_7$	1	1	0	1	2
$\mathbf{o}_8$	0	0	0	0	2

Table 2: The observations $\mathbf{o}_i$ together with the binary attributes x_j and the associated label information y .

Solution 4. Using the Bayes rule we can compute the probability as follows

$$p(y = 2 | x_2 = 1, x_4 = 1) = \frac{p(x_2 = 1, x_4 = 1 | y = 2)p(y = 2)}{p(x_2 = 1, x_4 = 1 | y = 1)p(y = 1) + p(x_2 = 1, x_4 = 1 | y = 2)p(y = 2)}$$

and we find the associated probabilities from the given table e.g. how many times the event $x_2 = 1, x_4 = 1$ appears when $y = 2$.

Question 5. Which one of the following statements about logistic regression is *correct*?

- A. The attributes of a logistic regression model cannot be modified in order to construct classifiers with nonlinear decision boundaries.
- B. In terms of a bias-variance trade-off, a logistic regression model with a well-tuned regularization parameter has a negligible bias but a fairly high variance.
- C. Consider a regularized logistic regression classifier. For a fixed value of λ , there exists a closed-form solution for the parameters $\mathbf{w}$, obtained by differentiating the objective function and setting the derivative to zero.
- D. Consider a regularized logistic regression model for a binary classification problem with a balanced dataset. If the estimated test error is approximately 0.5, this typically indicates over-regularization.**
- E. Don't know.

Solution 5.

- We can use nonlinear transformations of the attributes to get nonlinear decision boundaries.

- If the regularization parameter is well-tuned we expect the bias and the variance to be relatively low.
- There is no closed-form optimal solution for the logistic regression problem and we need to optimize the parameters using numerical optimization.
- An over-regularized logistic regression model will push all weights to zero except for the bias term. Hence, the trained model will always predict as label the class with the highest number of training points. Consequently, if we estimate the generalization error, we expect all the points of the relatively smaller class to be misclassified, so the error will be approximately 50% as the dataset is balanced.

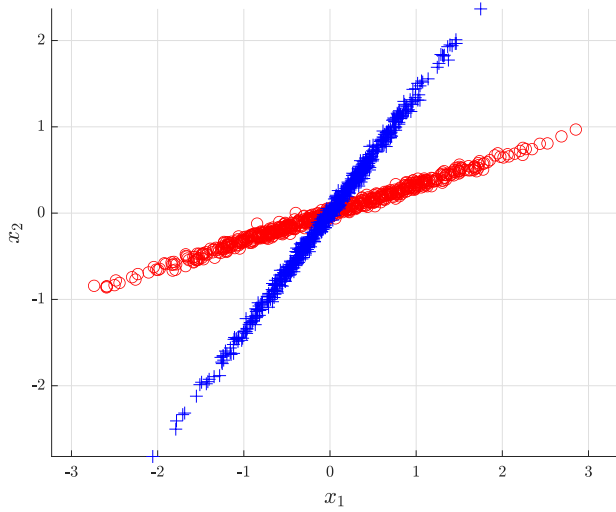


Figure 3: A two dimensional dataset of two classes indicated by blue (+) markers and red (o) markers.

Question 6. Suppose a PCA analysis is carried out on a two dimensional dataset presented in Figure 3 where two classes can be distinguished. In Figure 4 the projections of the dataset onto four different directions are presented, one per panel. Two of these plots correspond to the projection on the first and the second principal component.

Which one of the options below is *correct*?

- A. First principal direction corresponds to Plot 3
Second principal direction corresponds to Plot 4
- B. First principal direction corresponds to Plot 1
Second principal direction corresponds to Plot 2**
- C. First principal direction corresponds to Plot 1
Second principal direction corresponds to Plot 2
- D. First principal direction corresponds to Plot 2
Second principal direction corresponds to Plot 1
- E. Don't know.

Solution 6.

The correct solution is B, check the PCA direction in Figure 5.

Question 7. We consider a binary classification problem where the input has two attributes $\mathbf{x} = [x_1, x_2]^\top$. We model the class conditional distributions with the

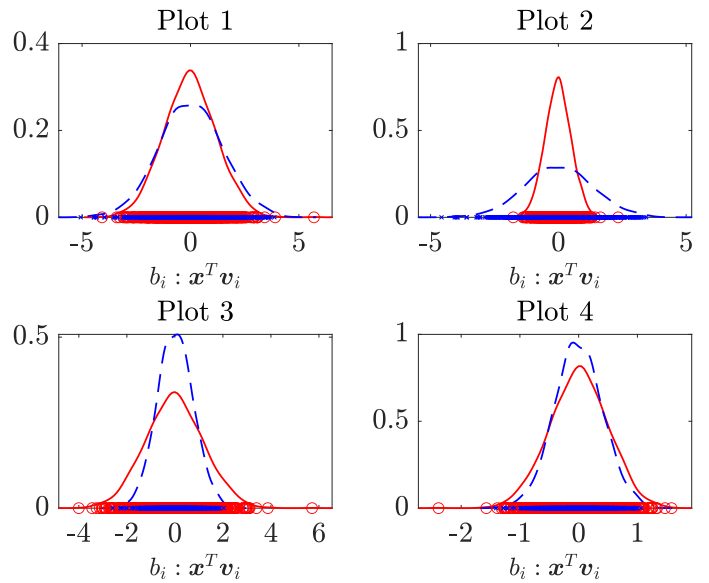


Figure 4: The projection of the dataset onto four different vectors. Note that each class is represented with a different density in each plot. In particular, the red class with marker (o) is represented by the continuous density curve, and the blue class with marker (+) is represented by the dashed density curve.

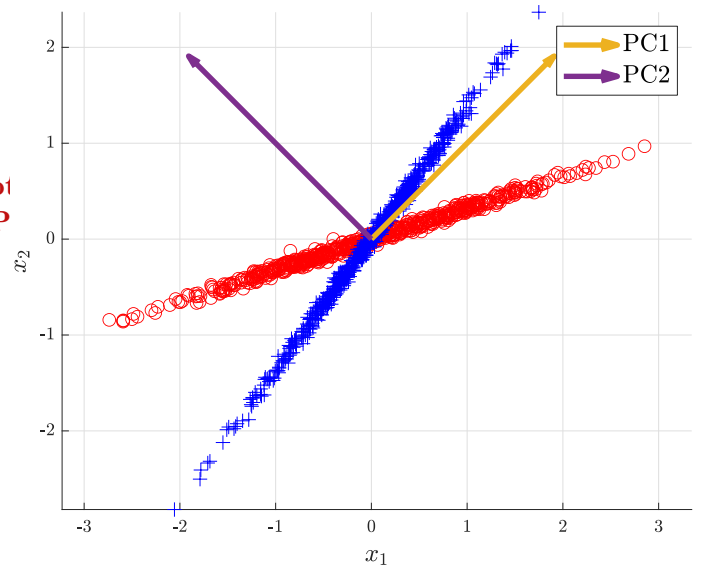


Figure 5: SOLUTION: The PCA directions. The first principal direction captures most of the variance of the dataset, so the associated projections will have the highest variance. The second principal direction is orthogonal to the first and the distribution of the projections follows.

following multivariate normal distributions

$$p(\mathbf{x} \mid y = j) = \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}^{(j)}, \boldsymbol{\Sigma}^{(j)})$$

$$\boldsymbol{\mu}^{(j)} = \begin{bmatrix} \mu_1^{(j)} \\ \mu_2^{(j)} \end{bmatrix}, \quad \boldsymbol{\Sigma}^{(j)} = \begin{bmatrix} \Sigma_{11}^{(j)} & \Sigma_{12}^{(j)} \\ \Sigma_{21}^{(j)} & \Sigma_{22}^{(j)} \end{bmatrix}, \quad \text{for } j = 1, 2.$$

We use the *Naive Bayes* approach to derive a decision rule for classifying a point $\mathbf{x}$ in *Class 1*.

Which one of the following statements is *correct*?

Hint: Consider the ratio of the probabilities.

A. $\frac{(x_1 - \mu_1^{(2)})^2}{\Sigma_{11}^{(2)}} + \frac{(x_2 - \mu_2^{(2)})^2}{\Sigma_{22}^{(2)}} - \frac{(x_1 - \mu_1^{(1)})^2}{\Sigma_{11}^{(1)}} - \frac{(x_2 - \mu_2^{(1)})^2}{\Sigma_{22}^{(1)}} + \log \Sigma_{11}^{(2)} + \log \Sigma_{22}^{(2)} - \log \Sigma_{11}^{(1)} - \log \Sigma_{22}^{(1)} > 2 \log \frac{p(y=2)}{p(y=1)}$

B. $\frac{(x_1 - \mu_1^{(2)})^2}{\Sigma_{11}^{(2)}} + \frac{(x_2 - \mu_2^{(2)})^2}{\Sigma_{22}^{(2)}} - \frac{(x_1 - \mu_1^{(1)})^2}{\Sigma_{11}^{(1)}} - \frac{(x_2 - \mu_2^{(1)})^2}{\Sigma_{22}^{(1)}} + \log \Sigma_{11}^{(2)} + \log \Sigma_{22}^{(2)} - \log \Sigma_{11}^{(1)} - \log \Sigma_{22}^{(1)} > 2 \log \frac{p(y=1)}{p(y=2)}$

C. $(\mathbf{x} - \boldsymbol{\mu}^{(2)})^\top (\boldsymbol{\Sigma}^{(2)})^{-1} (\mathbf{x} - \boldsymbol{\mu}^{(2)}) + \log |\boldsymbol{\Sigma}^{(2)}| - (\mathbf{x} - \boldsymbol{\mu}^{(1)})^\top (\boldsymbol{\Sigma}^{(1)})^{-1} (\mathbf{x} - \boldsymbol{\mu}^{(1)}) - \log |\boldsymbol{\Sigma}^{(1)}| > 2 \log \frac{p(y=2)}{p(y=1)}$

D. $(\mathbf{x} - \boldsymbol{\mu}^{(2)})^\top (\boldsymbol{\Sigma}^{(2)})^{-1} (\mathbf{x} - \boldsymbol{\mu}^{(2)}) + \log |\boldsymbol{\Sigma}^{(2)}| - (\mathbf{x} - \boldsymbol{\mu}^{(1)})^\top (\boldsymbol{\Sigma}^{(1)})^{-1} (\mathbf{x} - \boldsymbol{\mu}^{(1)}) - \log |\boldsymbol{\Sigma}^{(1)}| > 2 \log \frac{p(y=1)}{p(y=2)}$

E. Don't know.

Solution 7. A new observation $\mathbf{x}_* = [x_1, x_2]^\top$ will be classified in the *Class 1* if the condition $p(y = 1 | x_1, x_2) > p(y = 2 | x_1, x_2)$ holds and by using the Naive Bayes assumption we compute:

$$\begin{aligned} p(y = 1 | x_1, x_2) &> p(y = 2 | x_1, x_2) \\ \Rightarrow \frac{p(x_1 | 1)p(x_2 | 1)p(1)}{\sum_j p(x_1 | j)p(x_2 | j)p(j)} &> \frac{p(x_1 | 2)p(x_2 | 2)p(2)}{\sum_j p(x_1 | j)p(x_2 | j)p(j)} \\ \Rightarrow \log[p(x_1 | 1)p(x_2 | 1)p(1)] &> \log[p(x_1 | 2)p(x_2 | 2)p(2)] \\ \Rightarrow -\frac{(x_1 - \mu_1^{(1)})^2}{2\Sigma_{11}^{(1)}} - \frac{1}{2} \log \Sigma_{11}^{(1)} + \dots &> \log \frac{p(y = 2)}{p(y = 1)}, \end{aligned}$$

and we can expand the following terms in a similar way to get the result.

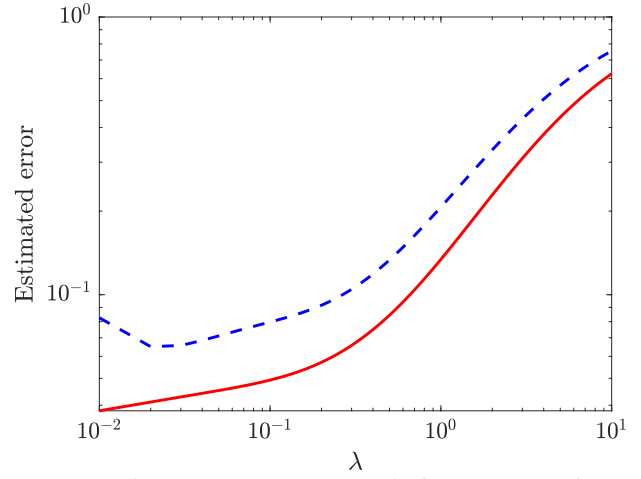


Figure 6: The training error and the estimated generalization error.

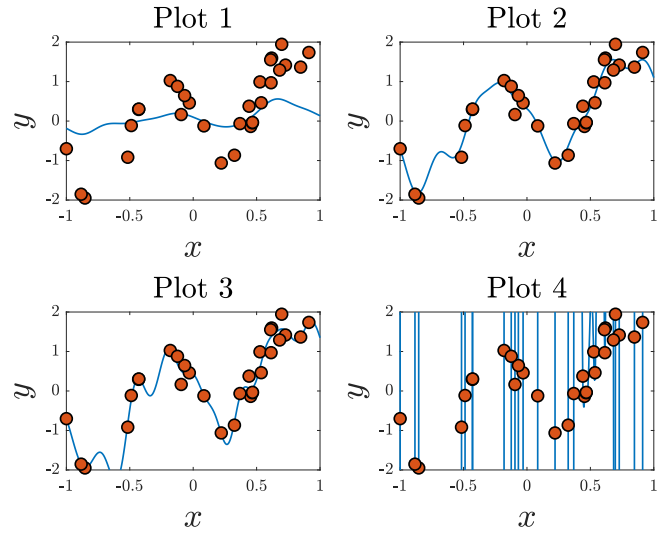


Figure 7: Each plot corresponds to the learned function $f(\mathbf{x}; \mathbf{w})$ shown as a blue curve for a different λ -value.

Question 8. We are given a dataset with N observations for a regression problem and we fit a regularized linear regression model $f(\mathbf{x}; \mathbf{w})$ with nonlinear transformation of the attributes and parameters $\mathbf{w}$. The cost function is written as

$$E(\mathbf{w}) = \frac{1}{N} \sum_{i=1}^N L(f(\mathbf{x}_i; \mathbf{w}), y_i) + \lambda \Omega(\mathbf{w}),$$

where the loss $L(f(\mathbf{x}; \mathbf{w}), y) \geq 0$ measures the distance between the true label y and the corresponding predicted value $f(\mathbf{x}; \mathbf{w})$, and the regularizer $\Omega(\mathbf{w}) \geq 0$ penalizes monotonically the weights i.e., the higher the weights, the higher the $\Omega(\mathbf{w})$. We apply K -fold cross-validation to estimate the generalization error, and the learning curves are shown in Figure 6 with the corresponding learned models in Figure 7.

Which one of the statements below is *correct*?

- A. Considering Figure 7, Plot 1 corresponds to low values of λ and Plot 4 to high values of λ .
- B. In Figure 6 the dashed blue curve represents the train error and the continuous red the test error.
- C. Plot 3 corresponds to a solution with higher variance compared to Plot 2.**
- D. When the cost function has the form as above, for any loss and regularizer, we can *always* compute the optimal weights as $\mathbf{w}^* = (\hat{\mathbf{X}}^\top \hat{\mathbf{X}})^{-1} \hat{\mathbf{X}}^\top \hat{\mathbf{y}}$.
- E. Don't know.

Solution 8.

- This is the correct answer as the function in Plot 3 is a bit more “wiggly” than the one in Plot 2, which indicates higher the bias in Plot 2.
- This formula does not hold always, is only valid when the loss function is the squared-loss and if we do not consider a regularization term ($\lambda = 0$).
- The behavior is exactly the opposite. When the λ is very high all the weights are going to 0 so the bias of the learned function is high, while for low values of λ the weights are not regularized which can cause the variance of the trained function to be high.
- The dashed blue represents the test error and the continuous red the train error as it always decreases which indicates overfitting.

Question 9. Which one of the statements is *correct*?

- A. For least squares linear regression the test error will always decrease as we include more nonlinear transformation of attributes in the model.
- B. Consider a dataset with N observations. Leave-one-out cross-validation will require fitting N models in total to find the best value of $\lambda \in [0, 10]$.
- C. Regularized linear regression always results in a better test error than unregularized linear regression regardless of the λ -value.
- D. Apart from regularization we can apply a bagging style approach to reduce the variance and increase the bias of a linear regression model with nonlinear transformation on the attributes.**
- E. Don't know.

Solution 9.

- No, it is typically the opposite as the complexity increases, as well as the risk to overfit.
- No, it requires fitting N models per value of λ .
- No, if the λ is too low/high we can overfit/underfit the data.
- Yes, as using different subsets of the data for training will give multiple models, perhaps with high variance each, but when computing the final model with averaging we reduce the individual effect and get a smoother final model.

Question 10. A regularized linear regression model is trained using the regularization term $\|\mathbf{w}\|_2^2$ with parameter $\lambda = 9$. We learn the parameters of the model based on the standardized training data $\hat{\mathbf{X}}$, where the mean and standard deviation of the training data are given by $\boldsymbol{\mu} = [1, 1]^\top$ and $\boldsymbol{\sigma} = [2, 6]^\top$, respectively. The learned parameters are $\mathbf{w}^* = [-6, -9]^\top$ and $w_0^* = 9$.

What is the output of the trained linear regression

model for a *new* test point $\mathbf{x}_{test} = [2, 3]^\top$?

- A. 1056
- B. 3**
- C. -30
- D. 1023
- E. Don't know.

Solution 10. We have first to standardize the new point and then apply the linear regression function so we have that

$$f(x) = \mathbf{w}^\top (\mathbf{x}_{test} - \boldsymbol{\mu}) / \sigma + b$$

where the division is done point-wise.

Model 1: Logistic Regression (LR)

		Predicted	
		Class 1	Class 2
Actual	Class 1	52	64
	Class 2	244	104

Model 2: Decision Tree (DT)

		Predicted	
		Class 1	Class 2
Actual	Class 1	60	56
	Class 2	110	238

Table 3: Confusion matrices.

Question 11. Table 3 shows the confusion matrices related to a Logistic Regression model (LR) and a Decision Tree classifier (DT) trained on a particular instance of the Earthquakes in Romania dataset to predict Earthquake level. The number of observations is $N = 464$. We define $y = 1$ to indicate a *positive outcome* and $y = 2$ to be a *negative outcome*.

Which one of the following statements is *correct*?

- A. Predicting everything as Negative gives worse accuracy than Model 1.
- B. Model 1 has higher recall than Model 2.
- C. Model 2 has higher precision than Model 1.**
- D. Model 1 has lower error rate than Model 2.
- E. Don't know.

Solution 11.

- Model 1 precision: 0.18, Model 2 precision: 0.35
- Model 1 recall: 0.45, Model 2 recall: 0.52
- Model 1 error rate: 0.66, Model 2 error rate: 0.36
- N: 464, Positive: 116, Negative: 348

Question 12. We consider a synthetic 1-dimensional dataset with the following five points:

$$X = \{-1, 5, 8, 12, 16\}$$

Suppose you want to perform k -means clustering with $K = 2$ centroids.

The centroids are initialized at $\mu_1 = 6$ and $\mu_2 = 11$. What will be the total cost, E , (i.e. the sum of squared distances from the observations to their nearest centroid) after the second iteration of k -means?

- A. 12
- B. 50
- C. 58
- D. 70
- E. Don't know.

Solution 12. The correct answer is B.

First iteration:

Allocation: [11122] Updated centroids:

$$\mu_1 = \frac{-1 + 5 + 8}{3} = 4$$

$$\mu_2 = \frac{28}{2} = 14$$

Nothing changes in the second iteration, i.e.

$$E = (-1-4)^2 + (5-4)^2 + (8-4)^2 + (12-14)^2 + (14-16)^2 = 50$$

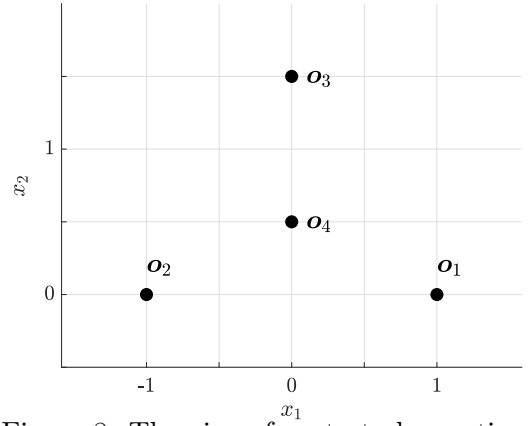


Figure 8: The given four test observations.

Question 13. A logistic regression model was trained on a two dimensional dataset using nonlinear transformations of the two attributes x_1 and x_2 such that for each observation $\mathbf{o} = [x_1, x_2]^\top$ the $p(y = 1 | \mathbf{o}) = \sigma([1, x_1^2, x_2^2]^\top \mathbf{w})$, where σ is the logistic sigmoid. Consider four test observations presented in Figure 8. We know for the associated weights $\mathbf{w} = [w_0, w_1, w_2]^\top$ that the condition $w_1 = w_2$ holds.

Additionally, the following probabilities are known:

$$p(y = 1 | \mathbf{o}_1) \approx 0.62, p(y = 1 | \mathbf{o}_2) \approx 0.62, \\ p(y = 1 | \mathbf{o}_3) \approx 0.85.$$

Which one of the following statements is *correct*?

Hint: The problem can be solved without complex computations; consider the shape of the decision boundary.

- A. $p(y = 1 | \mathbf{o}_4) > p(y = 1 | \mathbf{o}_1)$
- B. $p(y = 1 | \mathbf{o}_4) < p(y = 1 | \mathbf{o}_1)$
- C. $p(y = 1 | \mathbf{o}_4) = p(y = 1 | \mathbf{o}_3)$
- D. $p(y = 1 | \mathbf{o}_4) > p(y = 1 | \mathbf{o}_3)$
- E. Don't know.

Solution 13. As we observe in Figure 9 the correct solution is B. We could understand this by considering that $p(y = 1 | \mathbf{o}) = \sigma(\sigma([1, x_1^2, x_2^2]^\top \mathbf{w})) = \sigma(w_0 + w_1 \|\mathbf{o}\|^2)$ as $w_1 = w_2$ and which indicates that the higher the Euclidean distance $\|\mathbf{o}\|^2$ from 0, the higher the probability $y = 1$. Also, since the $p(y = 1 | \mathbf{o}_1) \neq p(y = 1 | \mathbf{o}_3)$ the w_1 cannot be 0.

Question 14. In a market basket problem, we consider an association rule $X \rightarrow Y$. Assume that the

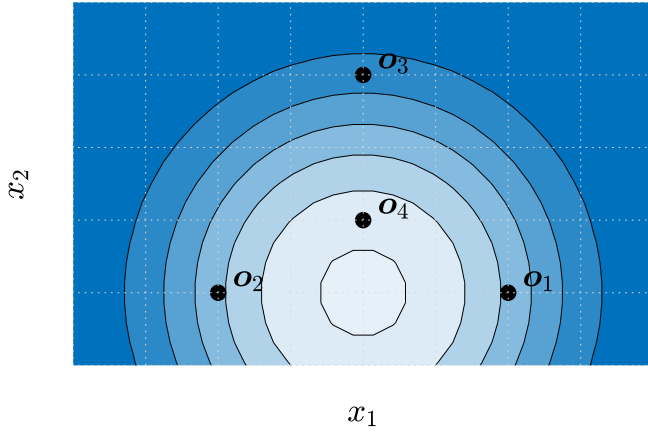


Figure 9: SOLUTION: The logistic regression classifier with the nonlinear transformation for the attributes.

support of $X \cup Y$ is $\frac{3}{10}$ and the confidence of $X \rightarrow Y$ is $\frac{4}{5}$. What is the support of X ?

- A. $support(X) = \frac{2}{10}$
- B. $support(X) = \frac{1}{2}$
- C. $support(X) = \frac{4}{10}$
- D. $support(X) = \frac{3}{8}$
- E. Don't know.

Solution 14. Using the definition of confidence, we have that

$$conf(X \rightarrow Y) = \frac{sup(X \cup Y)}{sup(X)}$$

$$sup(X) = \frac{sup(X \cup Y)}{conf(X \rightarrow Y)}$$

i.e.,

$$sup(X) = \frac{3/10}{4/5} = \frac{3/10}{8/10} = 3/8$$

I.e. essentially an applicaiotn of the product rule.

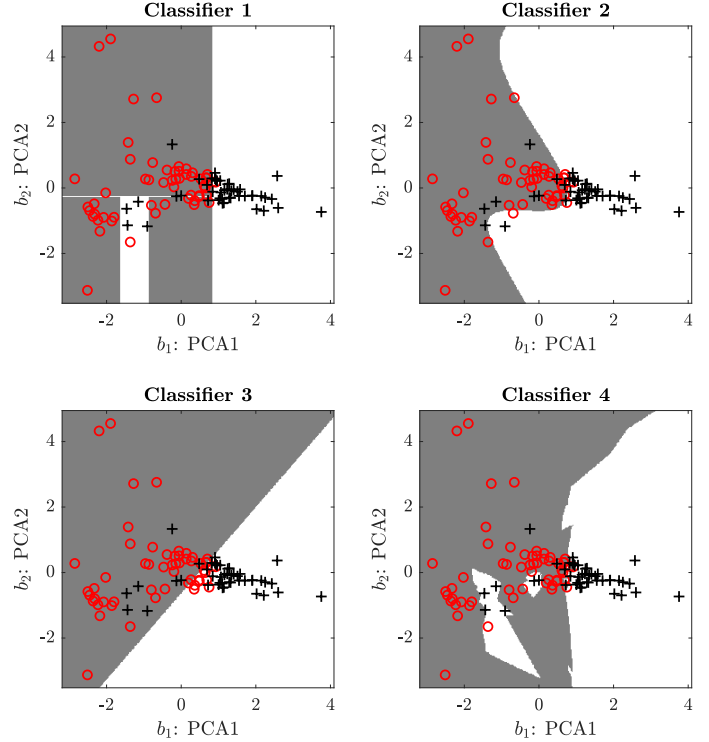


Figure 10: Decision boundaries for four different classifiers trained on the Earthquakes in Romania dataset when projected onto the first two principal components.

Question 15. We will consider a subset of the Earthquakes in Romania dataset (described in Table 1) after it has been projected onto the first two principal components b_1 and b_2 , thereby giving rise to a smaller two-dimensional dataset. We will consider the following four classifiers:

LREG: Logistic regression

ANN: Artificial neural network with 5 hidden units

DT: Decision tree with regular axis-aligned splits ($b_i < c$)

KNN: K-nearest neighbours with $K = 3$

Suppose the classifiers are trained on the two-dimensional dataset and the decision boundary for each of the four classifiers is given in Figure 10. Which one of the following statements is correct?

- A. Classifier 1 corresponds to **DT**,
Classifier 2 corresponds to **KNN**,
Classifier 3 corresponds to **ANN**,

Classifier 4 corresponds to **LREG**.

B. Classifier 1 corresponds to **DT**,

Classifier 2 corresponds to **LREG**,

Classifier 3 corresponds to **ANN**,

Classifier 4 corresponds to **KNN**.

**C. Classifier 1 corresponds to DT,
Classifier 2 corresponds to ANN,
Classifier 3 corresponds to LREG,
Classifier 4 corresponds to KNN.**

D. Classifier 1 corresponds to **KNN**,

Classifier 2 corresponds to **DT**,

Classifier 3 corresponds to **LREG**,

Classifier 4 corresponds to **ANN**.

E. Don't know.

Solution 15. To solve this problem, we have to use our intuition about what the typical decision boundaries for the different methods look like:

- A KNN method will have decision boundaries dictated by the nearest neighbors. That is, points (x, y) where the nearest K neighbors are in one class must be in the same class and therefore the boundaries will be fairly complex and respect the data distribution well.
- A decision tree has axis aligned splits, therefore the boundaries must be vertical or horizontal
- A logistic regression model must have linear boundaries
- An artificial neural network with few hidden units can have some non-linearity, but otherwise have boundaries of limited complexity and consisting of relatively simple shapes

It is easy to see this rules out all but option C.

Question 16. Which one of the following statements

about ensemble methods is *correct*?

- A. If the weighted error rate of a classifier in AdaBoost is $\varepsilon_t = 0.5$, then this classifier will have an influence on the result.
- B. In AdaBoost we can prevent overfitting when increasing the iterations if we use a regularized logistic regression classifier with fixed λ as the weak learner.
- C. In bagging and boosting observations are sampled without replacement, but in boosting misclassified observations are given a higher probability of being sampled in the next round.
- D. In AdaBoost, the more errors a classifier makes in a given round of boosting, the less the relative increase of the weights for the misclassified observations.**
- E. Don't know.

Solution 16.

- If the error of a classifier is $\varepsilon_t = 0.5$ then the associated $a_t = \frac{1}{2} \log \left(\frac{1-\varepsilon_t}{\varepsilon_t} \right) = 0$ which means that the classifier will not have any influence to the final prediction.
- Increasing the iterations in Adaboost will lead to overfitting even if the weak learner is a basic logistic regression model.
- In both approaches the observations are sampled *with* replacement.
- If the weighted error rate ε_t of a classifier increases the associated a_t decreases so the factor $\exp(a_t)$ will not increase a lot the new weights $\tilde{w}_i(t+1)$.

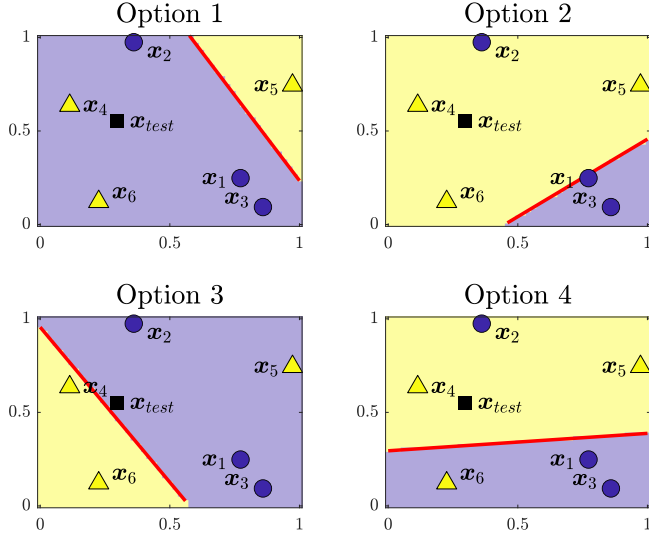


Figure 11: The four possible options for the classifier f_2 . The points of Class 0 are represented by $\bigcirc$ and of Class 1 by $\triangle$. The test point is represented by $\square$.

x	x_1	x_2	x_3	x_4	x_5	x_6
y	0	0	0	1	1	1
w_1	0.166	0.166	0.166	0.166	0.166	0.166
w_2	0.1	0.1	0.1	0.1	0.499	0.1

Table 4: The observations for the AdaBoost model.

Question 17. We consider the dataset in Figure 11 and the weights in Table 4 after two iterations of the AdaBoost algorithm. The predictions for a test point x_{test} for the two weak classifiers are $f_1(x_{test}) = 0$ and $f_2(x_{test}) = 1$. The final prediction using these two weak classifiers is $f^*(x_{test}) = 0$.

Which one of the options in Figure 11 corresponds to the classifier f_2 ?

Hint: Consider the weighted error rates of f_1 and f_2 .

- A. Option 1
- B. Option 2
- C. Option 3
- D. Option 4**
- E. Don't know.

Solution 17. Given the information of the question:

- We realize that the error of the second classifier must be higher than the error of the first classifier. This is the case because the final prediction is equal to the prediction of f_1 and the prediction of f_2 is not taken into account.

Outer fold $i = 1$

Inner folds	$j = 1$	$j = 2$	$j = 3$	Test
Model 1	1	9	4	55
Model 2	6	4	19	43

Outer fold $i = 2$

Inner folds	$j = 1$	$j = 2$	$j = 3$	Test
Model 1	18	4	8	47
Model 2	15	4	9	29

Table 5: The information for the two-level cross-validation. For each outer fold, we are given the number of misclassified points for each model for each inner fold on the associated validation set, together with the number of misclassified points on the associated test partition of the outer fold.

- We know that the $f_2(x_*) = 1$ which excludes automatically some options.
- We know that the error of f_1 is represented implicitly by the number of points that got a higher weight w_2 , and the error for each given f_2 in Figure 11 can be computed by summing the w_2 for the points that were misclassified.

Only for option D the prediction $f_2(x_*) = 1$ while the indices of the misclassified points for f_1 are $[[0], [0], [0], [0], [1], [0]]$ and for f_2 are $[[0], [1], [0], [0], [0], [1]]$ which gives $\varepsilon_2 = 0.2 > 0.17 = \varepsilon_1$.

Question 18. We consider a subset of $N = 120$ points from the Earthquakes in Romania dataset and two different classifiers (Model 1 and Model 2). We apply two-level cross-validation to estimate the generalization error, with 2 outer folds and 3 inner folds. In Table 5 the information for each outer fold, model, and inner fold is given, together with the performance of each model on the test partition of the outer fold.

Which one of the following options is a correct estimate of the generalization error?

- A. $\hat{E}^{\text{gen}} = 0.6$
- B. $\hat{E}^{\text{gen}} = 0.7$**
- C. $\hat{E}^{\text{gen}} = 0.85$
- D. $\hat{E}^{\text{gen}} = 0.35$
- E. Don't know.

Solution 18.

As the number of outer-folds is 2, the size of each test fold is $N_{test} = 120/2$. So when consider the test error of each selected model we need to divide the number of misclassified observations by N_{test} . Then we compute the average of the resulting test errors.

Question 19. Which one of the following statements about cross-validation (CV) is *correct*?

- A. Ideally, the same test set should be used for estimating the parameters of the model and for the final evaluation of the model's performance.
- B. In K -fold cross-validation the trained models are less similar to each other compared to those in leave-one-out cross-validation because the overlap between the training sets is smaller.**
- C. Two-fold cross-validation is the same as the hold-out method when 50% is hold-out.
- D. The cross-validation technique is *exclusively* used for supervised learning problems to estimate the generalization error.
- E. Don't know.

Solution 19.

- Ideally, the model selection and the evaluation of the model's performance should be done using independent test sets to avoid spurious correlations.
- This is true as there is a smaller overlap between the training sets that are used during the K -fold procedure.
- In hold-out method only 1 model is trained and is then tested, while in the 2-fold cross-validation two models are trained and then tested.
- We can use cross-validation in unsupervised learning as well, for example to estimate the optimal number of components in a Gaussian mixture model, by using the negative-log-likelihood as the "error" measure.

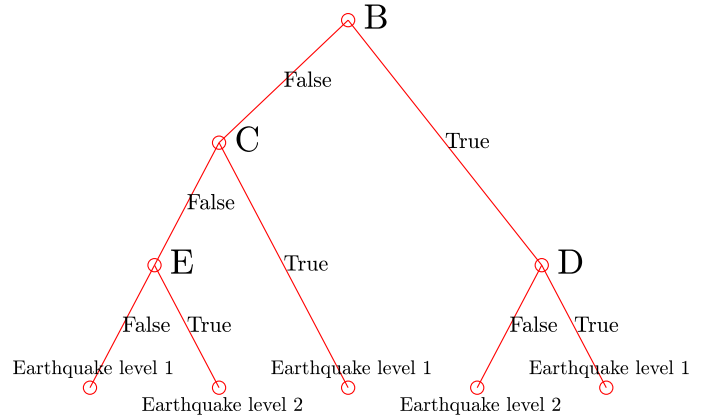


Figure 12: Structure of decision tree. The goal is to determine the splitting rules.

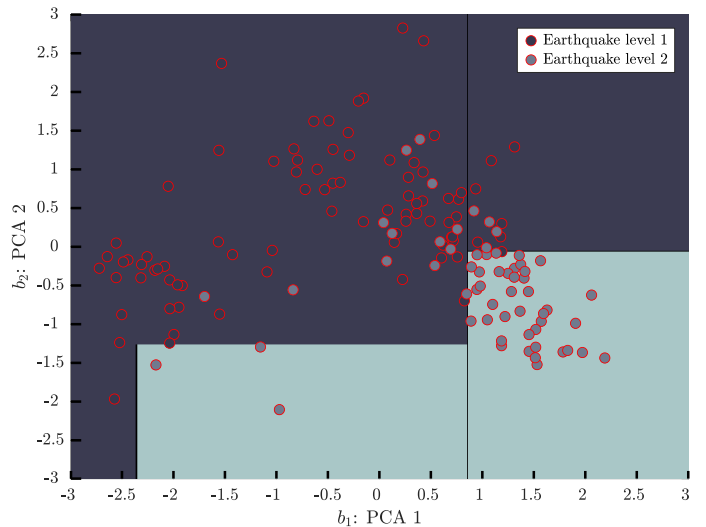


Figure 13: Classification boundary.

Question 20. We consider the Earthquakes in Romania dataset projected onto the first two principal directions. Suppose we train a decision tree to predict which of the two classes an observation belongs to. Since the attributes are continuous, we will consider binary splits of the form $b_i \geq z$ for different values of i and z , where b_1, b_2 refer to the coordinates of the observations when projected onto principal directions. Suppose the trained decision tree has the form shown in Figure 12, and that according to the tree the predicted label assignment for a subset of the observations is as given in Figure 13, what is then the correct rule

assignment to the nodes in the decision tree?

- A. **B**: $b_2 \geq -1.27$, **C**: $b_2 \geq -0.07$, **D**: $b_1 \geq -2.38$, **E**: $b_1 \geq 0.83$
- B. **B**: $b_1 \geq 0.83$, **C**: $b_2 \geq -0.07$, **D**: $b_2 \geq -1.27$, **E**: $b_1 \geq -2.38$
- C**. **B**: $b_1 \geq 0.83$, **C**: $b_2 \geq -1.27$, **D**: $b_2 \geq -0.07$, **E**: $b_1 \geq -2.38$
- D. **B**: $b_2 \geq -0.07$, **C**: $b_1 \geq -2.38$, **D**: $b_2 \geq -1.27$, **E**: $b_1 \geq 0.83$
- E. Don't know.

Solution 20.

This problem is solved by using the definition of a decision tree and observing what classification rule each of the assignment of features to node names in the decision tree will result in. I.e. beginning at the top of the tree, check if the condition assigned to the node is met and proceed along the true or false leg of the tree.

The resulting decision boundaries for each of the options are shown in Figure 14 and it follows answer C is correct.

Question 21. Consider a two-dimensional dataset comprised of $N = 9$ observations shown in Figure 15. The dataset consists of three classes indicated by the blue squares (class 1), red triangles (class 2) and yellow circles (class 3). In the figure, the decision boundaries for four K -nearest neighbor classifiers (KNN) are shown. Which one of the plots correspond to the $K = 3$ nearest-neighbour classifier assuming ties are broken by assigning the point to the *nearest* neighbour's class?

- A. KNN decision boundary A**
- B. KNN decision boundary B
- C. KNN decision boundary C
- D. KNN decision boundary D
- E. Don't know.

Solution 21.

- KNN decision boundary A, is the correct solution with $K = 3$.
- KNN decision boundary D, here $K = 3$ but for the tie-break we use the label of the second nearest point.

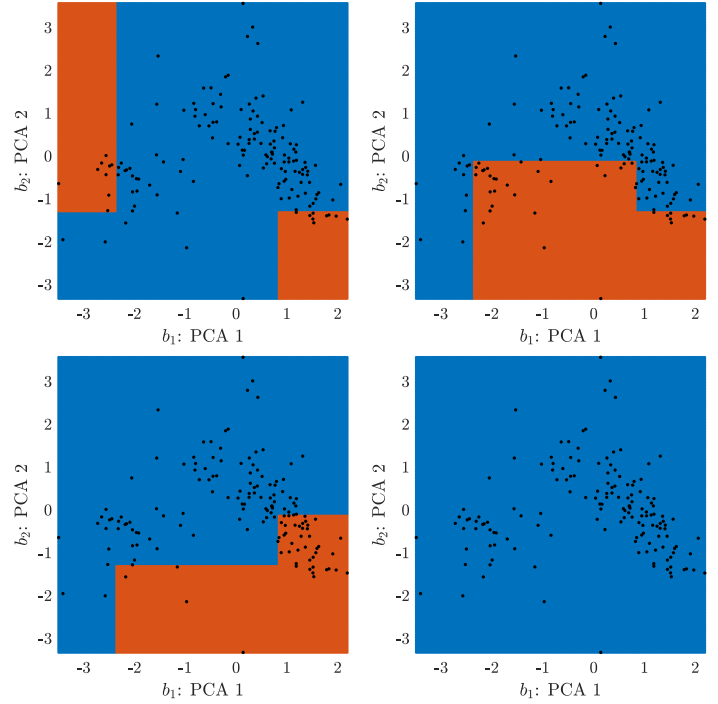


Figure 14: Classification trees induced by each of the options. (Top row: option A and B, bottom row: C and D)

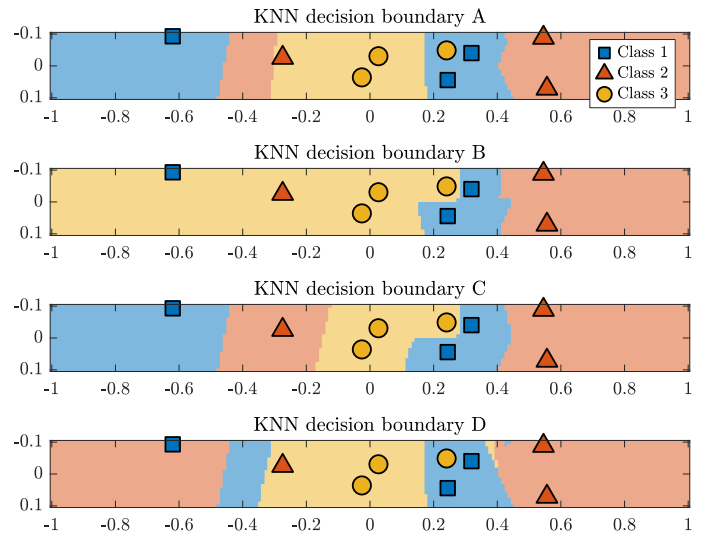


Figure 15: Decision boundaries for four KNN classifiers.

- KNN decision boundary B, here $K = 4$.
- KNN decision boundary C, here $K = 1$.

Question 22. Which one of the following statements about artificial neural networks (ANNs) is *correct*?

- A. In a regression problem, an ANN with linear activation functions ($h^{(i)}(x) = x$) results in a complex model that tends to overfit if output depends on the input in a non-linear manner.
- B. Randomly initializing two ANNs with identical activation functions and number of hidden units, and training them with the same data, will always converge to the same model.
- C. The advantage of ANNs is that they require less data for training and are more interpretable and explainable compared to related techniques, such as linear regression.
- D. Increasing the number of hidden units in an ANN can lead to overfitting if not properly controlled through regularization.**
- E. Don't know.

Solution 22.

- An ANN with linear activation is equivalent to standard linear regression so can fit only linear data.
- Starting the training from two different starting points leads, in general, to different solutions as fitting an ANN is a non-convex problem (there is not a unique solution).
- ANNs are typically models with high complexity so they need a lot of data for training, and in addition it is hard to interpret and explain analytically their behavior.
- Increasing the number of hidden units means that the complexity of the model increases, which increases the risk of overfitting, especially if weights are not regularized.

Question 23. Consider the dendrogram from Figure 16. Suppose we apply a cutoff (indicated by the black line in Figure 16) thereby generating three clusters. We wish to compare the quality of this clustering, Q , to the ground-truth clustering, Z , indicated by the colors in Table 6.

Recall, $\mathbf{n}$ is the count matrix where element n_{km} indicates the number of observations assigned to cluster

	o_1	o_2	o_3	o_4	o_5	o_6	o_7	o_8	o_9	o_{10}
o_1	0.0	0.7	1.7	4.0	3.6	3.8	1.2	2.0	2.2	1.5
o_2	0.7	0.0	1.2	3.8	3.4	3.6	1.6	2.4	2.6	1.8
o_3	1.7	1.2	0.0	3.4	2.6	3.2	2.1	2.7	2.9	2.2
o_4	4.0	3.8	3.4	0.0	2.8	0.2	4.5	5.1	5.2	4.7
o_5	3.6	3.4	2.6	2.8	0.0	2.7	4.0	4.6	4.8	4.2
o_6	3.8	3.6	3.2	0.2	2.7	0.0	4.3	4.9	5.0	4.5
o_7	1.2	1.6	2.1	4.5	4.0	4.3	0.0	0.8	1.0	0.4
o_8	2.0	2.4	2.7	5.1	4.6	4.9	0.8	0.0	0.2	0.6
o_9	2.2	2.6	2.9	5.2	4.8	5.0	1.0	0.2	0.0	0.8
o_{10}	1.5	1.8	2.2	4.7	4.2	4.5	0.4	0.6	0.8	0.0

Table 6: The pairwise Euclidian distances, $d(o_i, o_j) = \|\mathbf{x}_i - \mathbf{x}_j\|_2 = \sqrt{\sum_{k=1}^M (x_{ik} - x_{jk})^2}$ between 10 observations from the Earthquakes in Romania dataset (recall that $M = 4$). Each observation o_i corresponds to a row of the data matrix $\mathbf{X}$ of Table 1. The colors indicate classes such that the black observations $\{o_1, o_2, o_3, o_4, o_5, o_6\}$ belong to class C_1 (corresponding to Low strength), and the red observations $\{o_7, o_8, o_9, o_{10}\}$ belong to class C_2 (corresponding to High strength). To avoid single features to dominate, the dataset was standardized by subtracting the mean and dividing by the standard deviation.

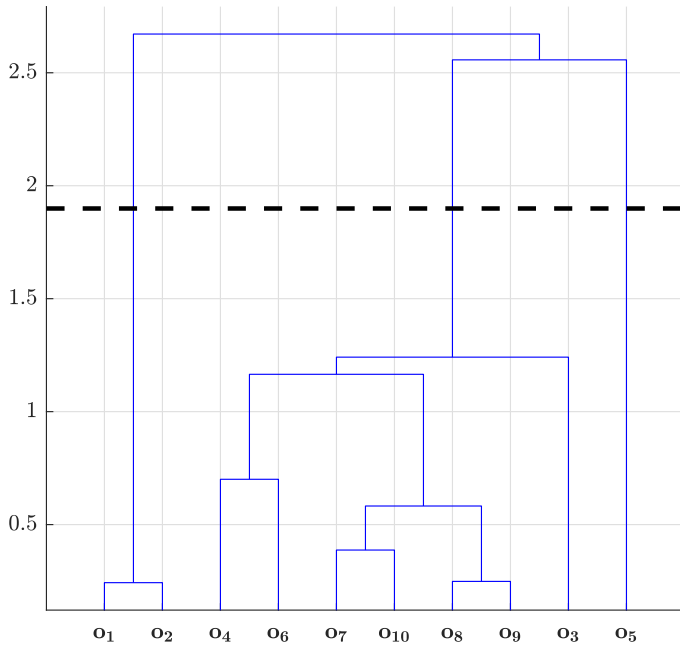


Figure 16: A hierarchical clustering is applied to the 10 observations in Table 6 (using unrounded values) with a cutoff indicated by the dotted line, thereby generating 3 clusters.

k in Z and m in Q . S is the number of times Z and Q agree that two observations are in the same cluster. D the number of times Z and Q agree that two observations are not in the same cluster.

Determine which one of the following set of intermediate calculations is correct.

A. $\mathbf{n} = \begin{bmatrix} 1 & 3 & 2 \\ 0 & 4 & 0 \end{bmatrix}$, $S = 10$, $D = 12$

B. $\mathbf{n} = \begin{bmatrix} 0 & 4 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 3 \end{bmatrix}$, $S = 9$, $D = 22$

C. $\mathbf{n} = \begin{bmatrix} 3 & 1 & 2 \\ 0 & 0 & 4 \end{bmatrix}$, $S = 10$, $D = 16$

D. $\mathbf{n} = \begin{bmatrix} 2 & 4 \\ 1 & 0 \\ 3 & 0 \end{bmatrix}$, $S = 10$, $D = 16$

E. Don't know.

Solution 23. The correct answer is .

We first identify Z from Table 6 noting the following allocation

$$Z = [1 \ 1 \ 1 \ 1 \ 1 \ 1 \ 2 \ 2 \ 2 \ 2]$$

. Q is identified by consider the dendrogram which gives

$$Q = [3 \ 3 \ 2 \ 2 \ 1 \ 2 \ 2 \ 2 \ 2 \ 2]$$

Note the column ordering is arbitray in Z and Q .

Elements in $\mathbf{n}$, n_{km} , indicates the number of observations assigned to cluster k in Z and m in Q , which gives

$$\begin{bmatrix} 1 & 3 & 2 \\ 0 & 4 & 0 \end{bmatrix}$$

$$\mathbf{n}^Z = [1 \ 7 \ 2]$$

$$\mathbf{n}^Q = [6 \ 4]$$

Which gives $S = 10$ and $D = 12$ from equation 18.13 and 18.17 in the lecture notes.

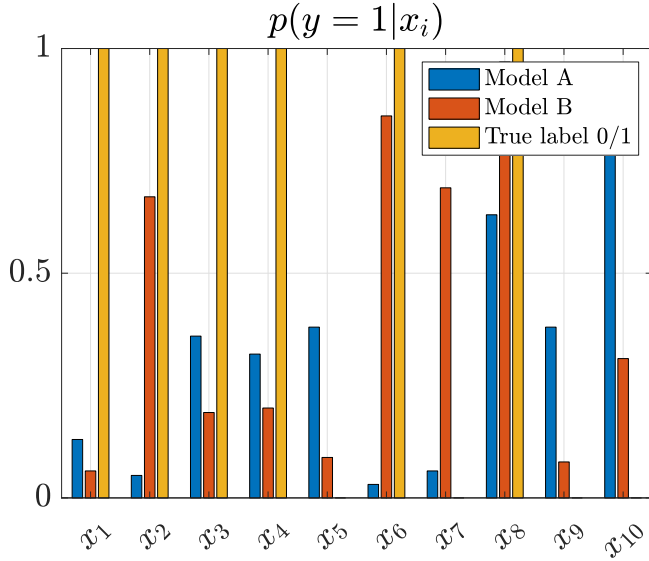


Figure 17: The $p(y = 1|x_i)$ for the two models on a test set of 10 points $x_1, \dots, x_{10}$, and the true test labels. If $p(y = 1|x_i) > 0.5$, then x_i is classified to Class 1.

Question 24. We consider a statistical procedure to determine if there is a significant difference between two classifiers, Model A and Model B, using McNemar's test. The probabilities $p(y = 1 | x_i)$ for the two models on a test set of 10 points are shown in Figure 17, together with the true test label. Our null hypothesis is that the classifiers perform similarly.

Determine which combination of estimated difference in accuracy $\hat{\theta}$, confidence interval (CI) with $\alpha = 0.05$, and p-value is the only plausible answer.

- A. $\hat{\theta} = 0.4$, $CI = [-0.33, 1.0]$, p-value < 0.63
- B. $\hat{\theta} = -0.2$, $CI = [-0.54, 0.16]$, p-value > 0.5**
- C. $\hat{\theta} = 0.36$, $CI = [-0.54, 0.16]$, p-value < 0.63
- D. $\hat{\theta} = -0.2$, $CI = [-0.54, 0.16]$, p-value < 0.01
- E. Don't know.

Solution 24.

The only plausible option is B as the $\hat{\theta}$ is within the confidence interval (CI), the confidence interval is reasonable, also the $\hat{\theta}$ can be computed by the formula $E_{\theta} = \frac{n_{12} - n_{21}}{n}$, while a p-value $0.01 < \dots$ shows that two classifiers are extremely different (i.e., disagree in all cases), which is a contradiction if the CI includes 0 and the $\hat{\theta}$ is in the CI.

Question 25. A neural network classifier is trained to

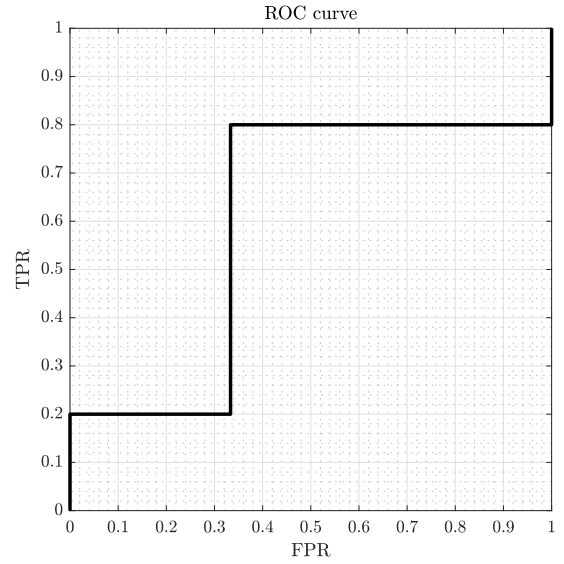


Figure 18: ROC curve for a neural network classifier, where the predictions and true class labels are one of the options in Figure 19.

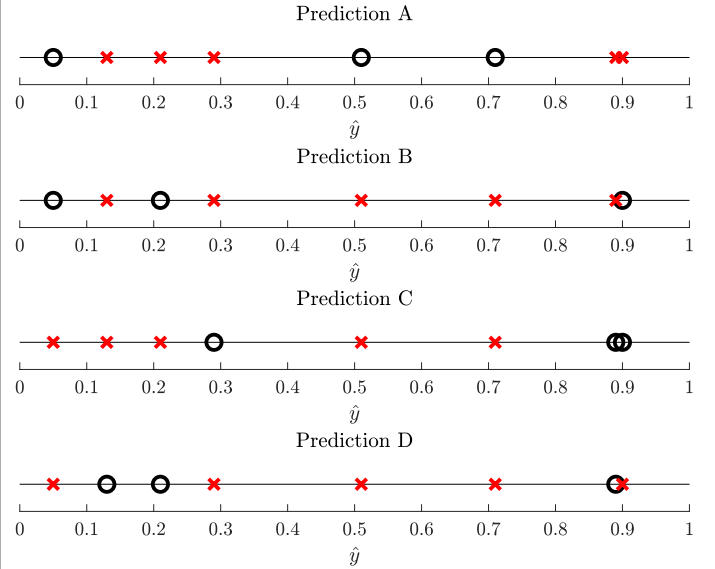


Figure 19: Four candidate predictions for the ROC curve in Figure 18. The observations are plotted horizontally, such that the position on the x -axis indicate the predicted probability $\hat{y} = p(y = 2|x)$, and the marker/color indicate the class membership, such that the black circles indicate the observation belongs to the negative class $y_i = 1$ and red crosses to the positive class $y_i = 2$.

distinguish between two classes $y \in \{1, 2\}$ and produce class-probability $\hat{y}$ and the *receiver operator characteristic* (ROC) curve of the network when evaluated on a test set with $N = 8$ observations is shown in Figure 19. Suppose we plot the predictions on the $N = 8$ test observations by their $\hat{y}$ value along the x -axis and indicate the class labels by either a black circle (negative class $y = 1$) or red cross (positive class $y = 2$), which one of the subplots in Figure 19 then corresponds to the ROC curve in Figure 18?

- A. Prediction A
- B. Prediction B
- C. Prediction C
- D. Prediction D**
- E. Don't know.

Solution 25. The correct answer is D. To see this, recall that the ROC curve is computed from the false positive rate (FPR) and true positive rate (TPR) for particular choices of threshold value $\hat{y}$. To compute e.g. the TPR, one assumes every observation predicted to belong to class 1 with a probability higher than $\hat{y}$ is actually assigned to class one. We then divide the total number of observations belonging to class one *and which are predicted to belong to class 1* with the number of observations in the *positive* class.

Similarly for the FPR, where we now count the number of observations that are assigned to class one *but in fact belongs to class 0*, divided by the total number of observations in the *negative* class.

This procedure is then repeated for different threshold values to obtain the curves shown in Figure 20. The ROC curve is then obtained by plotting these two curves against each other. I.e. for each threshold value, the point

$$(x, y) = (\text{FPR}, \text{TPR})$$

is on the AUC curve. This rules out all options except D. For completeness, we have included the ROC curves for all options in Figure 21.

Question 26. We consider an artificial neural network with the following first layer weights $\mathbf{w}_1^{(1)} = [-3, -1]^\top$ and $b_1^{(1)} = 3$ and $\mathbf{w}_2^{(1)} = [1, 0]^\top$ and $b_2^{(1)} = 2$, and the second layer $\mathbf{w}^{(2)} = [-2, -2]^\top$ and $b^{(2)} = 1$. We consider the ReLU activation $h^{(1)}(x) = \max\{0, x\}$ in the first layer and the identity activation in the second

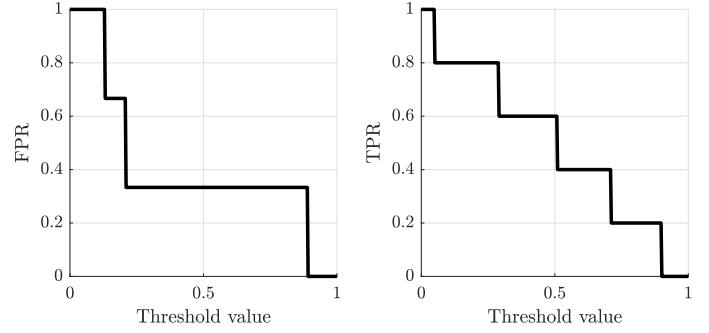


Figure 20: TPR, FPR curves for the classifier.

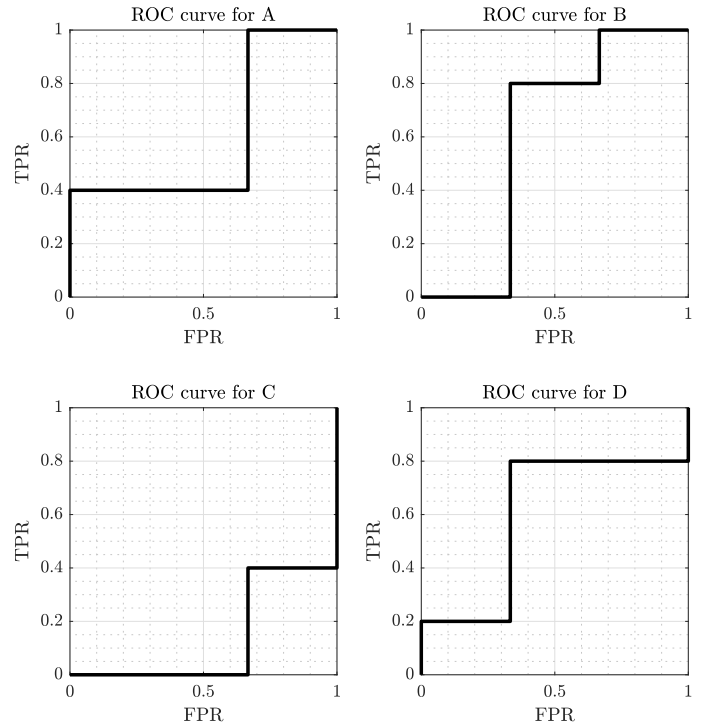


Figure 21: ROC curves for all options.

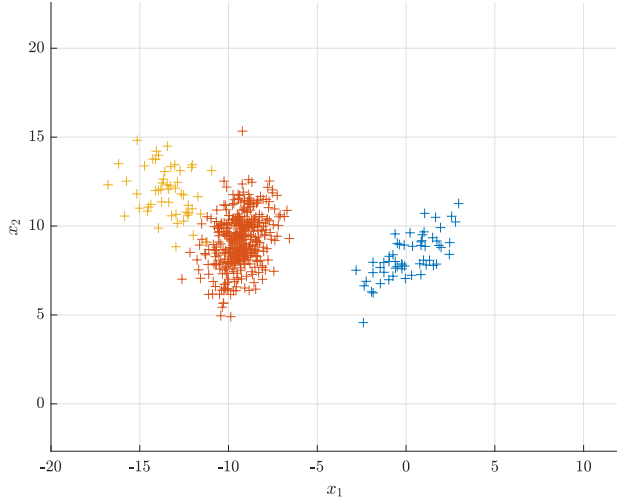


Figure 22: 500 observations drawn from a Gaussian Mixture Model (GMM) with three clusters.

layer $h^{(2)}(x) = x$, and a test point $\mathbf{x}_{test} = [3, 3]^\top$. What is the correct output of the neural network?

- A. $f(\mathbf{x}_{test}; \mathbf{w}) = -9$
- B. $f(\mathbf{x}_{test}; \mathbf{w}) = 9$
- C. $f(\mathbf{x}_{test}; \mathbf{w}) = 1$
- D. $f(\mathbf{x}_{test}; \mathbf{w}) = -27$
- E. Don't know.

Solution 26. The formula to evaluate the neural network is

$$f(x) = h^{(2)} \left(\sum_i h^{(1)} \left(x_{test}^\top \mathbf{w}_i^{(1)} + b_i^{(1)} \right) w_i^{(2)} + b^{(2)} \right)$$

where $h^{(2)}(x) = x$ and $h^{(1)}(x) = \max\{0, x\}$, which gives the result.

Question 27. Let $\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma})$ denote the multivariate normal distribution with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$. In Figure 22 is given 500 observations drawn from a density defined by a Gaussian Mixture Model (GMM) with three clusters. Each observation is colored in terms of which cluster it came from in the Gaussian Mixture model.

Which one of the following GMM densities was used

to generate the data?

A.

$$p(\mathbf{x}) = \frac{1}{10} \mathcal{N} \left(\mathbf{x} \mid \begin{bmatrix} 0.2 \\ 8.2 \end{bmatrix}, \begin{bmatrix} 1.4 & -0.4 \\ -0.4 & 2.0 \end{bmatrix} \right) + \frac{4}{5} \mathcal{N} \left(\mathbf{x} \mid \begin{bmatrix} -9.3 \\ 9.3 \end{bmatrix}, \begin{bmatrix} 2.4 & 1.0 \\ 1.0 & 1.4 \end{bmatrix} \right) + \frac{1}{10} \mathcal{N} \left(\mathbf{x} \mid \begin{bmatrix} -13.3 \\ 11.8 \end{bmatrix}, \begin{bmatrix} 1.0 & 0.4 \\ 0.4 & 2.5 \end{bmatrix} \right)$$

B.

$$p(\mathbf{x}) = \frac{4}{5} \mathcal{N} \left(\mathbf{x} \mid \begin{bmatrix} 0.2 \\ 8.2 \end{bmatrix}, \begin{bmatrix} 1.0 & 0.4 \\ 0.4 & 2.5 \end{bmatrix} \right) + \frac{1}{10} \mathcal{N} \left(\mathbf{x} \mid \begin{bmatrix} -9.3 \\ 9.3 \end{bmatrix}, \begin{bmatrix} 2.4 & 1.0 \\ 1.0 & 1.4 \end{bmatrix} \right) + \frac{1}{10} \mathcal{N} \left(\mathbf{x} \mid \begin{bmatrix} -13.3 \\ 11.8 \end{bmatrix}, \begin{bmatrix} 1.4 & -0.4 \\ -0.4 & 2.0 \end{bmatrix} \right)$$

C.

$$p(\mathbf{x}) = \frac{1}{10} \mathcal{N} \left(\mathbf{x} \mid \begin{bmatrix} 0.2 \\ 8.2 \end{bmatrix}, \begin{bmatrix} 2.4 & 1.0 \\ 1.0 & 1.4 \end{bmatrix} \right) + \frac{4}{5} \mathcal{N} \left(\mathbf{x} \mid \begin{bmatrix} -9.3 \\ 9.3 \end{bmatrix}, \begin{bmatrix} 1.0 & 0.4 \\ 0.4 & 2.5 \end{bmatrix} \right) + \frac{1}{10} \mathcal{N} \left(\mathbf{x} \mid \begin{bmatrix} -13.3 \\ 11.8 \end{bmatrix}, \begin{bmatrix} 1.4 & -0.4 \\ -0.4 & 2.0 \end{bmatrix} \right)$$

D.

$$p(\mathbf{x}) = \frac{1}{10} \mathcal{N} \left(\mathbf{x} \mid \begin{bmatrix} 0.2 \\ 8.2 \end{bmatrix}, \begin{bmatrix} 1.0 & 0.4 \\ 0.4 & 2.5 \end{bmatrix} \right) + \frac{1}{10} \mathcal{N} \left(\mathbf{x} \mid \begin{bmatrix} -9.3 \\ 9.3 \end{bmatrix}, \begin{bmatrix} 1.4 & -0.4 \\ -0.4 & 2.0 \end{bmatrix} \right) + \frac{4}{5} \mathcal{N} \left(\mathbf{x} \mid \begin{bmatrix} -13.3 \\ 11.8 \end{bmatrix}, \begin{bmatrix} 2.4 & 1.0 \\ 1.0 & 1.4 \end{bmatrix} \right)$$

E. Don't know.

Solution 27. The three components in the candidate GMM densities can be matched to the colored observations by their mean values. Then, by considering the basic properties of the covariance matrices, we can easily rule out all options except C. Alternatively, in Figure 23 is shown the densities for densities corresponding to option A (upper left), B (upper right) and D (bottom center).

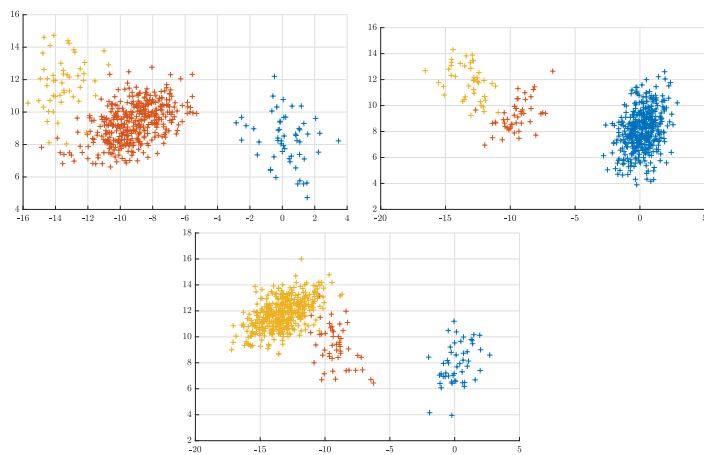


Figure 23: GMM mixtures corresponding to alternative options.